

4. Display: Why it matters

Your laptop's display is your window into your work (or play), so it's critical to get this right. Key specs include panel technology (OLED, IPS, mini-LED), resolution, refresh rate, brightness, and even more nuanced factors like PWM flicker and color gamut. Let's break down the differences and how to evaluate a screen, including some step-by-step tests you can do:



■ **OLED vs IPS vs mini-LED:** These refer to the panel technology:

- **IPS (In-Plane Switching)** is the most common LCD type in laptops. Pros: Good colour and wide viewing angles (no major color shift if you tilt the screen), generally uniform and reliable. Cons: Blacks are actually dark gray (since there's a backlight always on) – in a dark room, you might see light glow in black areas. Standard IPS without local dimming also can't reach the very high contrast or HDR performance of the other types. However, high-end IPS displays can still be excellent, some covering wide color gamuts or reaching 500 nits brightness. They also don't suffer from burn-in and usually don't use aggressive PWM for brightness (so less eye-strain risk).
- **OLED (Organic LED)** panels have each pixel emit its own light. This means "true black" (pixels shut off = zero brightness, infinite contrast) and very vivid colors. OLEDs often cover 100% DCI-P3 color gamut (which is ~25% more colors than sRGB) easily, giving you punchy, saturated visuals great for media and design. They also typically have very fast pixel response times (no motion blur) – useful for gaming. Cons: Potential burn-in – static UI elements (taskbars, etc.) can very slowly wear out OLED pixels if displayed at high brightness for thousands of hours. Modern OLED laptops mitigate this with pixel shifting and other tricks, so burn-in is rare for typical usage, but it's a consideration if you keep the same app open 24/7. Also, OLED displays can be less bright for full-screen white. They excel at HDR with small highlights (often peaking 500–600 nits on a window, some even 1000 nits on very small areas) but if you display a full white screen (say, editing a Word doc), many OLEDs will only do ~250–300 nits to avoid overheating. In contrast,

an IPS can often sustain its max (e.g. 400 nits) across the whole screen. So in bright sunlight, OLED's advantage diminishes. Another con: Many OLED laptops use PWM to dim (essentially flickering the pixels on and off rapidly to simulate lower brightness). If the PWM frequency is low, some people sensitive to flicker may get eye strain or headaches. High-quality OLEDs drive PWM into high kHz ranges to mitigate this. Still, if you have sensitive eyes, you might prefer IPS which often is DC-dimmed or flickers less.

- **Mini-LED** is a newer tech that's kind of the best of both worlds between IPS and OLED. It's essentially an LCD panel (often IPS-type) but with a backlight made of many tiny LEDs divided into local dimming zones. This can yield excellent contrast – for example, in dark scenes, portions of the backlight turn off, approaching OLED-level blacks (though not per-pixel, so there can be some halo/bloom around bright objects on black backgrounds). Mini-LED laptops (like the Apple MacBook Pro 14/16 with Liquid Retina XDR, or some ASUS ProArt StudioBooks) can reach very high brightness (1000+ nits for HDR highlights) while still offering wide color (often full DCI-P3) and no risk of burn-in. Cons: Some blooming artifacts (a slight glow around small bright details on black background), and these panels are usually in very pricey models. Also, mini-LED displays often use PWM for dimming the zones, but at high frequencies. Manufacturers are improving zone algorithms to minimize blooming. If you watch a lot of HDR content or work in HDR content creation, a mini-LED or high-end OLED is worth considering. Otherwise, a good IPS is typically sufficient and often more affordable.

- **Resolution & size:** Common laptop resolutions in 2025: 1920×1080 (FHD), 2560×1440 or 2560×1600 (QHD/QHD+), 3840×2160 (4K), and some in-between (3K, 2.8K, etc.). On a 13–14 inch, FHD is okay but QHD is crisper for text; 4K is often overkill (and drains battery), though some 13–14" OLEDs are 2.8K or 3K which strike a nice balance of sharpness. On 15–16 inch screens, QHD (around 1440p) is a sweet spot; 4K is gorgeous for photo/video work (and necessary if you're doing 4K video editing to see pixel-level detail), but it will use more power and can make UI elements tiny (Windows scaling helps, but some older apps don't scale well). For general use, we recommend at least a 1080p screen; avoid any 1366×768 panels lingering on very low-end 14" laptops – they look noticeably pixelated. If you can get 1440p or higher on a 14–16", you'll enjoy crisper text and more screen real estate. Just remember higher resolution = more GPU load, so it can lower battery life a bit and, if gaming, you'll get fewer FPS at native resolution (though you can always set games to 1080p).
- **Refresh rate:** Standard is 60Hz. Many laptops now offer 90Hz, 120Hz, or even 144Hz+ panels. Higher refresh makes motion smoother – you'll notice it in fast scrolling, anima-